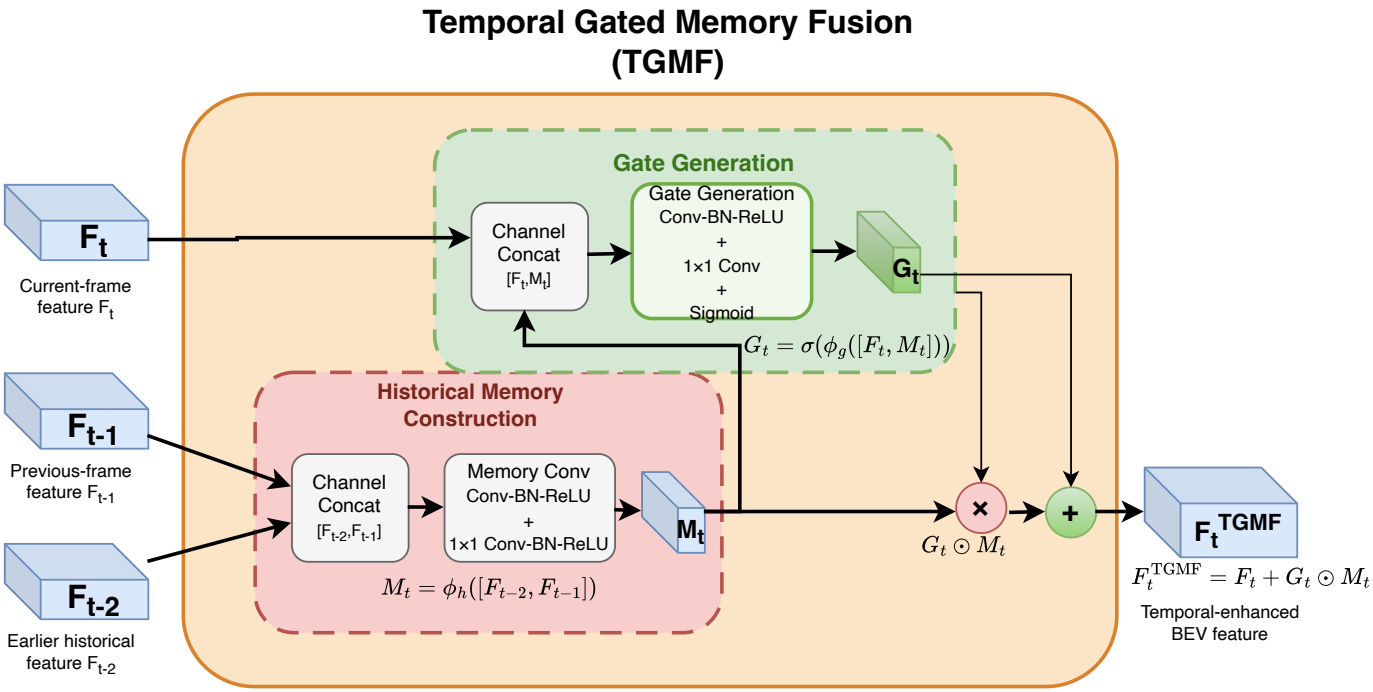

Temporal Gated Memory Fusion
(TGMF)
Gate Generation
Channel
Concat
[Ft,Mt]
Gate Generation
Conv-BN-ReLU
+
1×1 Conv
+
Sigmoid
Gt
$G_t = \sigma(\phi_g([F_t, M_t]))$
Ft
Current-frame
feature Ft
Ft-1
Previous-frame
feature Ft-1
Ft-2
Earlier historical
feature Ft-2
Historical Memory
Construction
Channel
Concat
[Ft-2,Ft-1]
Memory Conv
Conv-BN-ReLU
+
1×1 Conv-BN-ReLU
Mt
$M_t = \phi_h([F_{t-2}, F_{t-1}])$
×
$G_t \odot M_t$
+
Ft TGMF
$F_t^{\mathrm{TGMF}} = F_t + G_t \odot M_t$
Temporal-enhanced
BEV feature